U1A
1V8
SWIN
SW_EN
3V3_RF
3V3_RF
MUX_3V3_1V8
NVCC_SD2
SWOUT
1V8
1V8
1V8
SD_VSELECT
VSYS2
VSYS2
VSYS2
PMIC_STANDBY
3V3
VSYS2
VSYS2
VSYS2
3V3
3V3
VSYS
VSYS
VSYS
3V3
3V3
CLKIN1
CLKIN2
+1V8
NVCC_SD2
SD_VSELECT
+5V
+3V3
U1C
ENET1_MDC
ENET1_MDIO
ENET1_RXD3
ENET1_RXD2
ENET1_RXD1
ENET1_RXD0
ENET1_RX_CTL
ENET1_RXC
ENET1_TXD3
ENET1_TXD2
ENET1_TXD1
ENET1_TXD0
ENET1_TX_CTL
ENET1_TXC
U1D
ENET2_TX_CTL
ENET2_TXC
ENET2_TXD3
ENET2_RXD3
ENET2_RXD2
ENET2_TXD2
ENET2_TXD0
ENET2_RX_CTL
ENET2_TXD1
ENET2_RXD0
ENET2_RXC
ENET2_RXD1
U1F
PCM_MCLK
SAI3_MCLK
SAI3_TXFS
SAI3_TXD
SAI3_RXD
SAI3_TXC
PCM_OUT
PCM_SYNC
PCM_CLK
PCM_IN
U1I
RF_ANT_EXT
IW612_JTAG_TDI
IW612_JTAG_TMS
IW612_JTAG_TCK
WL_DEV_WAKE
BT_DEV_WAKE
IW612_JTAG_TDO
BT_HOST_WAKE
WL_HOST_WAKE
J1
IMU_BOARD
+3V3
ICM_SCLK
ICM_MOSI
ICM_CS
ICM_INT1
BMI_SCLK
BMI_MOSI
BMI_SDO
BMI_GYR_CS
BMI_ACC_INT
GND
ICM_MISO
ICM_INT2
BMI_ACC_CS
BMI_GYR_INT
U1E
ConnectCore93_LGA
USB_OTG1_N
USB_OTG1_P
USB_OTG2_N
USB_OTG2_P
USB1_VBUS
USB_OTG1_ID
USB_OTG2_ID
WLAN_SD3_DATA3
WLAN_SD3_DATA2
WLAN_SD3_DATA1
WLAN_SD3_DATA0
WLAN_SD3_CMD
WLAN_SD3_CLK
SD2_RESET_B
USB2_VBUS
SD2_CMD
SD2_DATA0
SD2_CD_B
SD2_DATA1
SD2_DATA2
SD2_DATA3
SD2_CLK
SD2_VSELECT
U1G
SPI1_CS0
SPI1_SIN
SPI8_SCK
SPI1_SOUT
BT_UART1_RXD
BT_UART1_TXD
SPI1_SCK
SPI8_SOUT
SPI8_CS0
SPI8_SIN
I2C4_SDA
15.4_SPI_FRM
15.4_SPI_INT
15.4_SPI_RXD
15.4_SPI_TXD
15.4_SPI_CLK
BT_UART1_RTS
BT_UART1_CTS
I2C4_SCL
UART7_CTS
UART7_RTS
UART7_TX
CAN2_TX
UART7_RX
CAN2_RX
CAN1_TX
CAN1_RX
I2C2_SCL
UART6_RX
I2C2_SDA
UART6_TX
I2C3_SDA
I2C3_SCL
ICM_SCLK
ICM_MOSI
ICM_CS
ICM_MISO
ENV_SDA
ENV_SCL
BMI_MOSI
BMI_SCLK
BMI_ACC_CS
PD_FAULT
BMI_SDO
PG_5V
OVP_FLT_N
GNSS_PPS
CHG_SCL
CHG_SDA
GNSS_RXD
GNSS_TXD
U1H
LVDS_D2_P
LVDS_D2_N
LVDS_D0_N
LVDS_D0_P
LVDS_CLK_P
LVDS_CLK_N
LVDS_D1_N
LVDS_D1_P
LVDS_D3_P
LVDS_D3_N
MIPI_CSI1_DATA1_P
MIPI_CSI1_DATA1_N
MIPI_CSI1_DATA0_P
MIPI_CSI1_DATA0_N
MIPI_CSI1_CLK_P
MIPI_CSI1_CLK_N
MIPI_DSI1_CLK_N
MIPI_DSI1_CLK_P
MIPI_DSI1_DATA0_N
MIPI_DSI1_DATA0_P
MIPI_DSI1_DATA1_N
MIPI_DSI1_DATA1_P
MIPI_DSI1_DATA2_N
MIPI_DSI1_DATA2_P
MIPI_DSI1_DATA3_N
MIPI_DSI1_DATA3_P
U1J
GPIO2_IO24
GPIO3_IO27(1V8)
GPIO2_IO19
GPIO2_IO23
GPIO2_IO22
TAMPER1(1V8)
TAMPER0(1V8)
ADC_IN0(1V8)
ADC_IN1(1V8)
GPIO1_IO10
GPIO2_IO06
GPIO4_IO14(1V8)
GPIO4_IO15(1V8)
ON_OFF
JTAG_TCK
JTAG_TDO
GPIO4_IO28(1V8)
ADC_IN3(1V8)
ADC_IN2(1V8)
GPIO4_IO29(1V8)
POR_B
JTAG_TDI
JTAG_TMS
PWR_ON
GPIO3_IO26(1V8)
GPIO2_IO18
SYS_RESET
BMI_GYR_CS
CHG_INT_N
ICM_INT2
ICM_INT1
BMI_GYR_INT
BMI_ACC_INT
ON_OFF_N
CHG_CE_N
U1B
U1K
GND
NC
RESERVED
R27
100k
ON_OFF_N
ON_OFF_SW
SW1
ON/OFF
C17
100nF
J2
USB-C PD / recovery
VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
U2
CYPD3177-24LQ
VBUS_MAX
VBUS_MIN
ISNK_COARSE
ISNK_FINE
VCCD
VBUS_IN
VDDD
FLIP
HPI_SCL
HPI_SDA
HPI_INT
VBUS_FET_EN
SAFE_PWR_EN
FAULT
GPIO_1
VDC_OUT
PD_FAULT
R3 5.1k
R4 1k
R5 5.1k
R6 1k
R7 5.1k
R8 5.1k
R12 10k CHG_SDA
R11 10k CHG_SCL
R13 10k CHG_INT_N
R1 100k CHG_CE_N
R2 3.04k CHG_PROG
C9 10uF
C6 47nF CHG_BTST1 CHG_SW1
L2 1.0uH CHG_SW1 CHG_SW2
R15 1k CHG_LED_A
D1 CHG CHG_STAT_N
C7 47nF CHG_BTST2 CHG_SW2
C10 10uF CHG_PMID
C8 4.7uF REGN
U3
BQ25798
VAC1
ACDRV1
VAC2
ACDRV2
SCL
SDA
INT
CE
QON
ILIM_HIZ
PROG
TS
PMID
BTST1
SW1
SW2
BTST2
STAT
REGN
SYS
BAT
SDRV
BATP
CHG_PROG
BAT_TS
VSYS_CHG
VBAT
C12 10uF
C11 10uF
R23 5.24k
R24 30.1k
J3
2S LiFePO4 6Ah + BMS
Q1
NMOS pass − TBD
OVP_HGATE
OVP_SENSE
+5V_RAW
R22 15m
C3 100nF BOOT_5V SW_5V
L1 4.7uH
C2 47uF
U4
LMR33630ADDA
VIN
EN
VCC
BOOT
SW
FB
PG
VCC_BUCK
FB_5V
PG_5V
R9 100k
R10 43.2k
C4 1uF
C1 10uF
R14 100k
U5
LTC4364CMS
HGATE
SOURCE
DGATE
SENSE
OUT
SHDN
UV
OV
TMR
ENOUT
FLT
OVP_FB
OVP_FLT_N
OVP_UV
OVP_OV
OVP_TMR
C5 0.22uF
R16 34k
R17 10k
R18 22.1k
R19 10k
R20 0.1
R21 10k
R25 4.7k ENV_SDA
R26 4.7k ENV_SCL
C13 100nF
C14 100nF
U6
BME690
CSB
SDI
SCK
SDO
VDD
VDDIO
U7
DAN-F10N
RXD
TXD
RESET_N
EXTINT
SAFEBOOT_N
TIMEPULSE
V_BCKP
VCC_RF
ANT_CTRL
EXT_RF_IN
LNA_EN
Reserved
GNSS_RXD
GNSS_TXD
GNSS_PPS
C15 10uF
C16 100nF

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